Assignment 7: Epistasis & Gene Linkage

Task 1

The goal of this work is to investigate the phenomenon of **epistasis** (non-linear interactions between genes) in evolutionary robotics, optimizing for *center-of-mass height* (`vertpos`) in creatures evolved under directional encoding (f_9) with the following 6-character directional stick encoding semantics (each indicates a direction towards which the respective stick grows from the parent (last added) stick).

Firstly, evolutionary optimization is run under the following parameters. Then 3 dissimilar individuals s_1, s_2, s_3 are selected from the final population $S^G, G = 300$ of evolved creatures as follows:

$$s_1 := \arg\max_{s \in S^G} \text{vertpos}$$

$$s_2 := \arg\max_{s \in S^G \setminus \{s_1\}} d(s, s_1)$$

$$s_3 := \arg\max_{s \in S^G \setminus \{s_1, s_2\}} \min\left(d(s, s_1), d(s, s_2)\right)$$

where $d(g_1, g_2)$ denotes weighted Normalized Levenshtein Distance and Directional Composition Divergence (see the script for the exact implementation).

Secondly, pairwise epistasis was estimated for the selected creatures based on the formula provided below.

Parameters & Setup

- **Fitness**: `vertpos`
- **Genetic Representation**: f_9
- **Simulation Environment (`eval-allcriteria.sim;deterministic.sim;sample-period-2.sim;only-body.sim`)**
- **Max parts**: `-max_numparts 30`
- **EA Parameters**:
  - **Population size**: `--popsize 100`
  - **Generations**: `--generations 300`
  - **Selection**: `--tournament 5`
  - **Crossover probability (`pxov`)**: `0.0`
  - **Mutation probability (`pmut`)**: `0.9`
  - **Hall of Fame size**: `-hof_size 1`
  - **Replications**: `24` independent runs run in parallel on 24 physical cores

Evolutionary Convergence & Performance

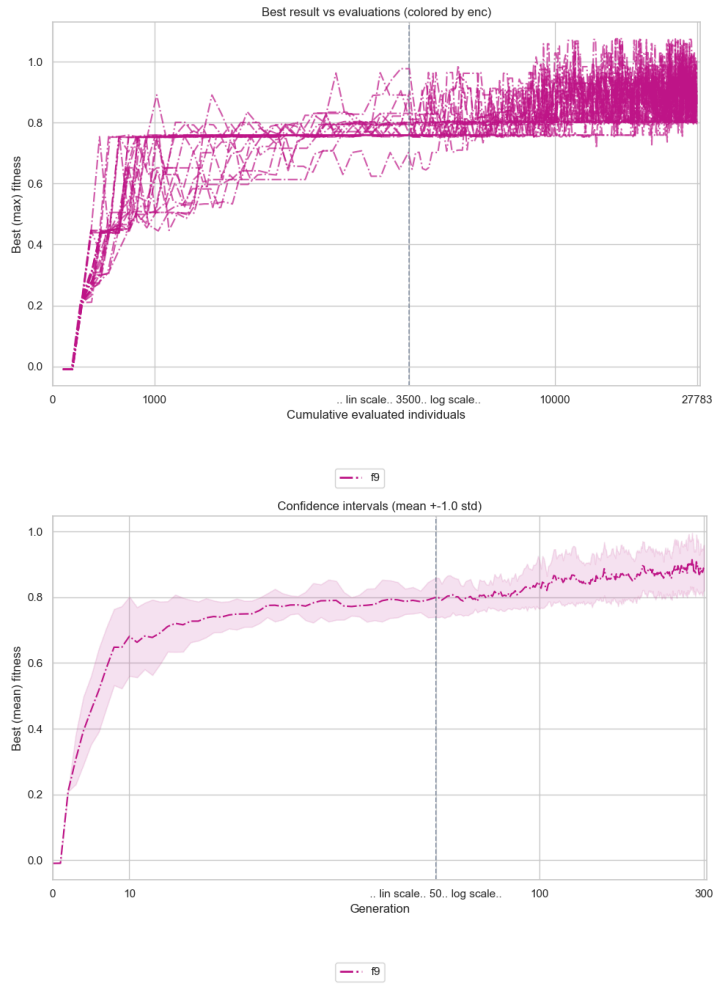


Figure 1a: *Hall of Fame best-fitness trajectories across 24 independent runs over 300 generations (left).* ***1b:*** *Mean best fitness across generations with confidence interval band (right).*

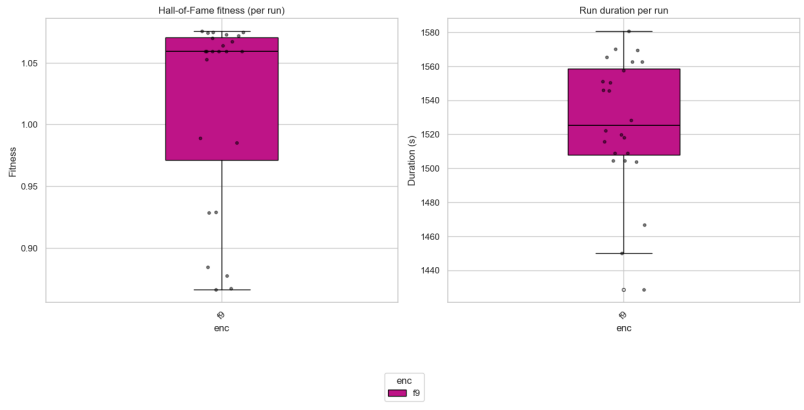


Figure 2: *Final Hall of Fame best fitness distribution and evaluation time across 24 replications.*

Epistasis Estimation Formula

For an intact genotype g of length L with baseline fitness $F(g)$:

- $F(g \setminus \{i\})$ - fitness after disabling gene i.
- $F(g \setminus \{j\})$ - fitness after disabling gene j.
- $F(g \setminus \{i,j\})$ - fitness after simultaneously disabling both genes i and j.

The additive null model predicts the joint knockout effect to be the sum of independent single knockouts (this holds iff genes are independent) $\Delta F_{ij} \equiv \Delta F_i + \Delta F_j$, where:

$$\Delta F_i = F(g \setminus \{i\}) - F(g), \qquad \Delta F_j = F(g \setminus \{j\}) - F(g)$$

For two genes $i \neq j$, pairwise epistasis $\varepsilon_{i,j}$ measures the interaction deviation from additivity:

$$\varepsilon_{i,j} = \underbrace{\left[F(g \setminus \{i,j\}) - F(g)\right]}_{\Delta F_{ij}} - \left(\Delta F_i + \Delta F_j\right) = F(g \setminus \{i,j\}) - F(g \setminus \{i\}) - F(g \setminus \{j\}) + F(g)$$

- $\varepsilon_{i,j} = 0 \implies$ additive / independent genes (White).
- $\varepsilon_{i,j} > 0 \implies$ positive epistasis (synergistic cooperation or functional redundancy / buffering - Green).
- $\varepsilon_{i,j} < 0 \implies$ negative epistasis (antagonistic breakdown / synthetic collapse - Red).

In the calculated matrices, the diagonal ($i = j$) stores the single-gene knockout effect $\Delta F_i (\equiv \Delta F_{ii})$.

Pairwise Epistasis Matrices (Task 1)

Using `select_dissimilar.py` (based on hybrid Levenshtein edit distance and 6D directional composition divergence), $k = 3$ dissimilar tall structures were selected from the evolutionary pool:

1. `/*9*/DDDDRLLRBBFUUUU` ($F = 1.0755$, $L = 15$) - top champion with vertical trunk ($UUUU$) on an asymmetric base.
2. `/*9*/FLLRFBLRRUUUUL` ($F = 0.9854$, $L = 14$) - wide-base tripod posture transitioning into a vertical mast.
3. `/*9*/DDDUDFRRBUUUDUF` ($F = 0.8776$, $L = 15$) - alternating stabilizing loops with upward segments.

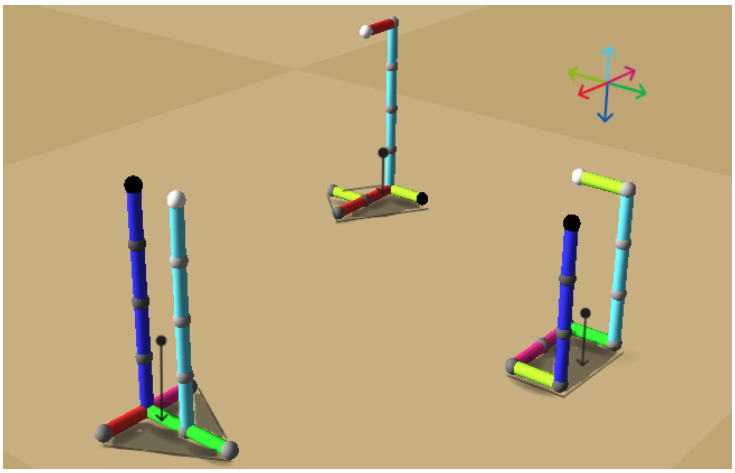


Figure 3: *Selected creatures (order preserved - left to right). All the creatures feature two main body parts: a plinthed base and a tall, erect spine (chord).* ***Center of mass*** *of each creature (black fat dots) is located above the base. Shaded in dark regions denote convex hulls formed by the support points (base sticks).* ***The criterion of static stability****: a projection of the center of mass onto the ground must lie strictly inside this convex hull (projections are indicated by black arrows).*

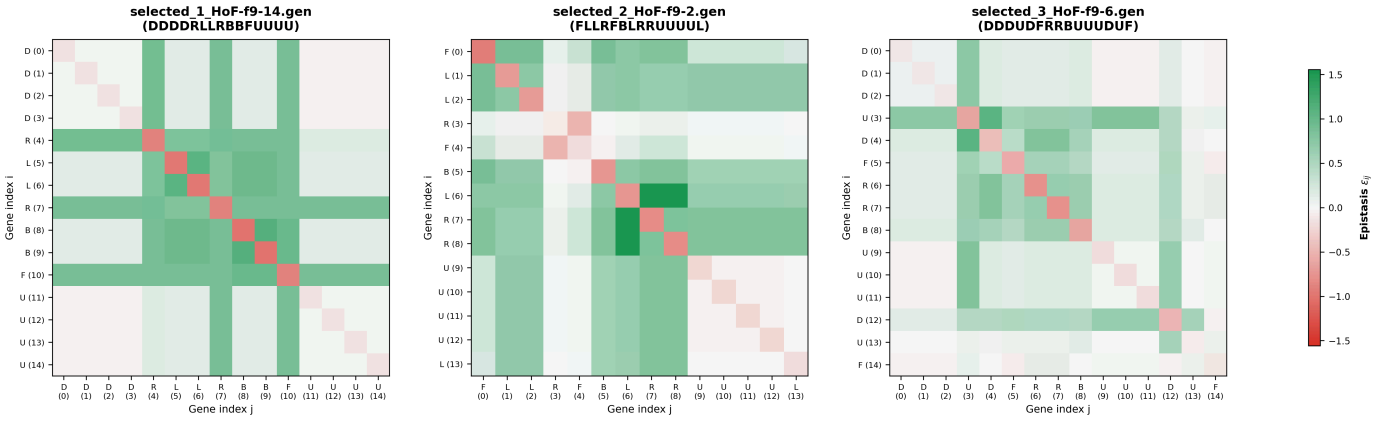


Figure 4: *Pairwise epistasis matrices for $k = 3$ dissimilar tall structures arranged in a grid. Axis labels display gene letter and positional index. Off-diagonal elements show epistatic interaction $\varepsilon_{i,j}$, while diagonal elements show single-gene knockout effects ΔF_i.*

Interpretation of individual effects and interactions

Phenotypic specialization - **Spine growth genes**

Creature 1 `/*9*/ DDDD R LL R BB F UUUU`	Creature 2 `/*9*/ F LL R F B L RR UUUU L`	Creature 3 `/*9*/ DDD U D F R R B UUU D U F`
0-3, 11-14. The first 4 `D` and the last 4 `U` genes correspond to the ***blue*** and ***cyan*** *twin-chords*, directly responsible for the **height** of the creature. Having **two (redundant) copies** slightly increases the center of the mass height (`vertpos`). Deleting these letters individually only slightly decreases `vertpos` (rows and columns 0 to 3 and 11 to 14 are mainly zero value).	**9-13**. Since these genes program growth of the spine (`UUUUL`) only (increasing `vertpos`), they do not interact with each other, but their positive (in terms of `vertpos`) effect depends on *base* genes (*0-3, 5-8*). In contrast to genes *9-12*, gene `L` serves stabilization purpose.	**0-3, 9-14**. Similar to the first (best) creature, there are 2 spines - ***blue*** and ***cyan***, encoded by the genes at the ends of genome. In contrast to the former creature, 2 genes - U^3 and D^{12} are the **key gene responsible for stability** of the creature, since both decrease long chords lengths (and hence two pairs of genes U^3D and $D^{12}U$ cancel out - the same effect described in the second creature) - without them the creature overturns. **4**. The role of the gene D^4 is complementary to the gene U^3, making it a **key gene responsible for morphological stability** - without this the ***blue*** chord would grow below *the base* level, overturning the creature.

Phenotypic specialization - **Plinth genes**

Creature 1	Creature 2	Creature 3
4. Gene `R` tells the direction to place the ***blue*** spine (`DDDD`) in the center of creature's plinth, critical to the stability of the creature - removing this gene places the center of mass allows small physical perturbations to overturn the creature, coloring the 4th row and column into *light green* (definitely positive effect). However, removal of just one `D` places the center of mass safely above the base. **5-6**. These genes are responsible for symmetry of a base, and hence are **critical for stability** of the whole creature (interaction with genes *4-10*). Genes encoding *spines* have little mutual effect with these two genes individually, the relationship will appear on *3D map of epistasis*. **7**. Gene `R` here is directly responsible for the symmetry of the base, placing the center of mass directly on the median segment of the base hull, yielding the highest "collaboration rate" (central *light green cross* on the matrix) - for instance, removing any other spinal gene restores the balance; removing any base gene does nothing - the creature already overturned (in both cases $\Delta F_{7,i} \gg \Delta F_7 + \Delta F_i$). **8-9**. Each of two genes `B` allows a ***cyan*** chord to appear - without one of these genes the second spine grows inside the first one and the creature falls down. **10** Gene `F` places the ***cyan*** spine towards the center of the base, which is critical for stability (without this gene a creature overturns).	**0-2**. Gene `F` allows for one of the supporting sticks to emerge (**key for stabilization**), genes `LL` move this 'leg' towards the only place, where the position of the creature is stable. **3-4**. These genes alter the base form but in the way that does not affect the stability of the creature, thus these genes show **little epistasis** (~zero rows/colums). **5-8**. These genes influence the shape of *the base* (`BL`) and position of *the spine* on it (`RR`), and thus have positive epistasis with spine-genes. Gene L^6 specifically has strong interaction with `RR` genes (*dense green*), shortening the base extended by 2 *R*-direction genes, thus **making the base hull compact**. Hence a pair `LR` cancels out (redundant subsequence)	**5-8**. Genes `FRRB` form a U-shaped *base*, placing the center of mass strictly inside *the base hull*. Each gene here has an individual interaction with every other base gene.

Specialization of genes extracted from pairwise epistasis matrix

The table above can be summarized as follows: after canceling out redundant subsequences of genes, rows and columns can be grouped in two groups:

- Groups that have noticeable interaction (*light green* to *dense green* squares) correspond to the *plinthed base* phenes (and have strong individual effects - *brighter red* diagonal elements)
- groups that has near-zero interaction with each other (`white` squares) and intermediate interactions (*pale green*) with the other genes, form *spine* phene groups. See Figure 4.

Individual effects (red diagonals)

Individual (diagonal) effects ΔF_i take negative values since extraction of any gene from the genome leads to a decrease in the fitness depending on the role of the gene:

- gene s encodes *spinal phene* $\implies$ height slightly decreases (*pale red* diagonal elements): $\Delta F_s \approx 0$
- gene b encodes a part of *base plinth* $\implies$ creature immediately falls down yielding near-zero fitness (*brighter red* diagonal elements): $\Delta F_b \ll 0$

Since removing any *base* gene overturns the creature, near-zero epistasis between *base* and *spinal* genes can be explained by individual effect of knocking out a base gene b

$$\varepsilon_{b,s} \approx 0 \iff \Delta F_{b,s} \approx \Delta F_b + \Delta F_s \approx \Delta F_b$$

In rare cases (base gene `7` in Creature 1, green cross on Figure 4) base and spinal genes strongly interact with each other.